Nitrogen losses from boreal arable soils after autumn manure application

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Abstract. We measured ammonia volatil-
isation, nitrous oxide emissions and nitrate leaching
on twelve clay and organic soil plots in western Finland over two years. Autumn application of cattle slurry increased nitrate leaching by 38 percent compared with spring application. Nitrous oxide emissions peaked during freeze-thaw cycles in early spring. Injection reduced ammonia losses by more than half compared with broadcast spreading.

1. Introduction. Agricultural nitrogen surplus remains a major source of nutrient loading to the Baltic Sea. Manure timing is regulated, but the evidence base for boreal conditions is thin. Previous studies were mostly conducted in temperate climates where soils rarely freeze. Snow cover and soil frost change the timing of microbial nitrogen transformations.

2. Methods. Each plot was equipped with suction cups at 30 and 60 cm depth and with static chambers for gas sampling. Ammonia volatilisation was measured with wind tunnels during the first 72 hours after application. Soil mineral nitrogen was sampled monthly. Weather data came from an on-site station.

3. Results. Nitrate concentrations in drainage water were highest in November and in April. On organic soils the leaching losses were nearly twice those of clay soils. Freeze-thaw cycles explained most of the annual nitrous oxide emission, while summer fluxes were small. The plots with slurry injection had the lowest ammonia losses in all weather conditions.

4. Discussion. Our results suggest that autumn application of manure should be avoided on organic soils in boreal regions. Delaying application to spring reduces leaching without lowering yields. Monitoring of soil frost could be used to time applications more precisely. Remote sensing of snow cover is a promising tool.

5. Conclusions. Spring application combined with injection minimised total nitrogen losses. Policy should account for soil type and frost conditions rather than calendar dates only.